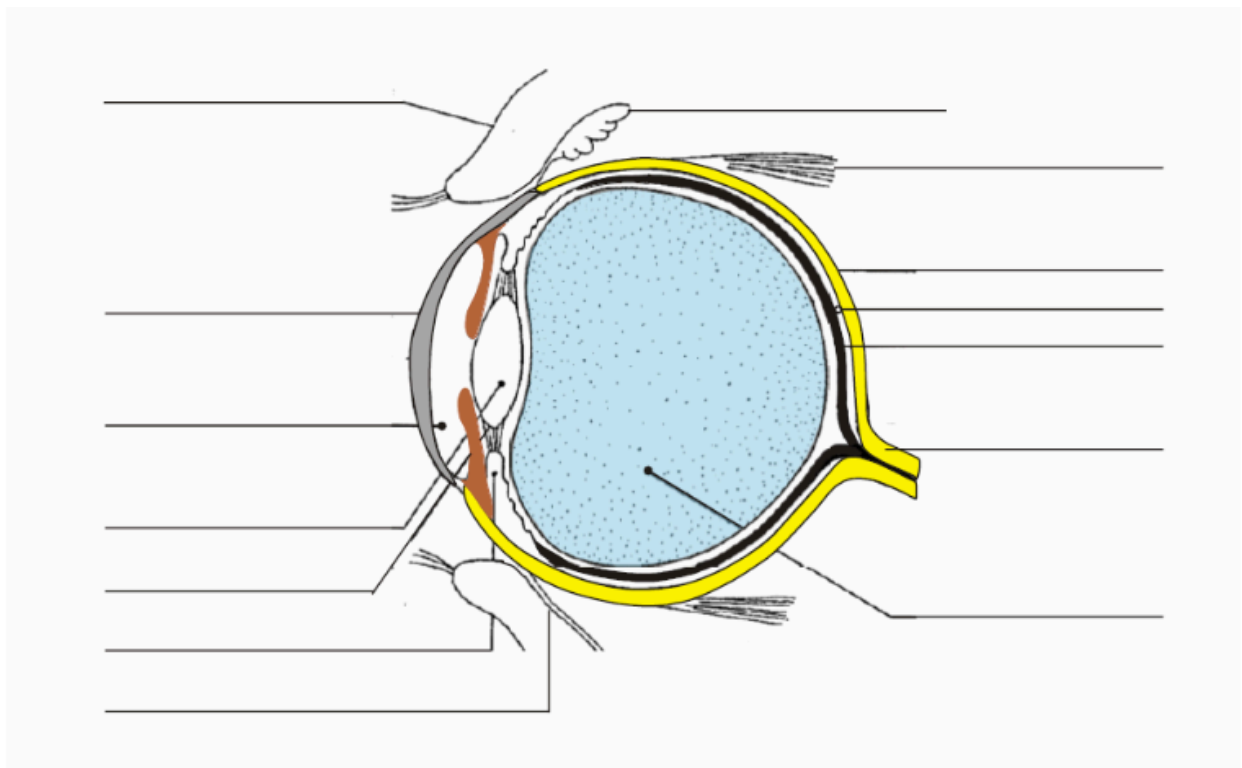


TOPIC 1.A

Structure, function, and protection of our sensory organs

1. Using the attached figure, describe the structure of the eye and the function of its individual parts.
2. Explain the accommodation of the eye to light conditions and to changes in the distance of the viewed object!
3. Interpret the refractive errors of the eye, and explain how and with what kind of glasses these can be corrected!
4. What is a cataract? What causes it? How is such a disorder treated?



Expected knowledge / Evaluation sheet	Attainable score	Achieved score
1. • The outer layer of the eye is the sclera • function: muscle attachment • and the transparent cornea. • function: refractive medium, protection. • Beneath the sclera is the choroid • function: blood supply, pigmentation • and the retina is found • function: generation of impulses in the receptors	8 points	
• The anterior part of the choroid is the iris, on which a circular opening, the pupil, is located. • Behind the pupil lies the lens, which is attached to the ciliary body / ciliary muscle by suspensory ligaments. • The interior of the eyeball is mostly filled by the vitreous humour, • the cavity behind the cornea up to the anterior limiting membrane of the vitreous humour is filled by the aqueous humour.	5 points	
2. • In bright light the pupil constricts / in dim light it	2 points	

dilates. • When viewing a distant object, the lens becomes flatter; cause: relaxation of the ciliary muscle. • When viewing a near object, the lens becomes more convex / the ciliary muscle contracts.	2 points	
3. • In the case of short-sightedness (myopia), the patient only sees well up close; the cause is a too-long eye axis / the refractive power of the eye is stronger than necessary (image forms in front of the retina). • Its correction can be achieved with a diverging lens (concave) / a lens with negative dioptries. • In the case of long-sightedness (hypermetropia), the patient does not see well up close / only sees well at a distance. • The cause is a too-short eye axis / the refractive power of the eye is weaker than necessary. • Its correction can be achieved with a converging lens (convex) / a lens with positive dioptries. 4. • cataract: partial or complete clouding of the lens, it becoming opaque. • surgically, a new, plastic lens can be inserted into the capsule of the diseased lens.	6 points 2 points	
5. Structure of the answer, communication skills	5 points	
Total:	30 points	

TOPIC 1.B

Characterisation of a plant species

1. Identify the received plant species using the Plant Identification book. Describe the process of identification.
2. To which major taxonomic groups does it belong? What are the characteristics of these groups that can be found on the received species?
3. Characterise the ecological requirements of the species.
4. In what type of habitat is the occurrence of this species typical? Where can we find such a place around the school?



Expected knowledge	Attainable score	Achieved score
1. Naming the species, listing the identification keys.	5 points	
2. Taxonomic group: phylum Angiosperms, class Dicotyledons • pinnate leaf venation, perennial, herbaceous • taproot system, characteristic flower, its leaves are opposite, covered with stinging hairs • its fruit is a nutlet	5 points	
3. Listing ecological requirements: moist, mesic (fresh) area; nutrient-rich, slightly calcareous soil; nitrogen indicator plant.	5 points	
4. Around the school, it occurs everywhere next to the fence in neglected (disturbed) areas (cosmopolitan species). Medicinal plant: used as an anti-inflammatory for joint problems, and in the case of difficulties with urination.	5 points	
Total:	20 points	